

第四章：原子的精细结构

§18 原子中电子轨道运动的磁矩

一、经典

载流线圈, $\vec{\mu} = iS\vec{e}_n$

$$\text{电子: } \vec{\mu} = iS\vec{e}_n = \frac{-e}{T} \pi r^2 \vec{e}_n \\ = -\frac{e v}{2\pi r} \pi r^2 = -\frac{e}{2m} \vec{L}$$

$$\text{记旋磁比 } \gamma = \frac{e}{2m} \text{ 则 } \vec{\mu} = -\gamma \vec{L}$$

在外场 $\vec{B}$ 中:

$$\text{力矩: } \frac{d\vec{L}}{dt} = \vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\Rightarrow \frac{d\vec{\mu}}{dt} = -\gamma \vec{\mu} \times \vec{B}$$

$\vec{\mu}$ 绕 $\vec{B}$ 以 $\omega = \gamma B$ 作进动

$$\text{进动频率 (拉莫尔频率): } \nu_L = \frac{\omega}{2\pi} = \frac{\gamma B}{2\pi}$$

二、量子

角动量: $L = \hbar \sqrt{l(l+1)}$

$$\text{磁矩: } \mu_L = -\gamma L = -\hbar \sqrt{l(l+1)} \frac{e\hbar}{2m_e}$$

z 方向角动量: $L_z = m_l \hbar$

$$\mu_{Lz} = -\gamma L_z = -\gamma m_l \hbar = -m_l \frac{e\hbar}{2m_e}$$

$$\text{记 } \mu_B = \frac{e\hbar}{2m_e} \text{ 玻尔磁子}$$

$\Rightarrow$ 磁量子数

$$\mu_L = -\hbar \sqrt{l(l+1)} \mu_B$$

$$\mu_{Lz} = -m_l \mu_B \quad m_l = 0, \pm 1, \dots, \pm l$$

$$\mu_B = \frac{1}{2} \frac{e^2 \hbar^2}{m_e e^2} \quad e\hbar = \frac{1}{2} \alpha c \hbar$$

$$\text{精细结构常数: } \alpha = \frac{e^2}{\hbar c}, \quad a_1 = \frac{\hbar^2}{m_e e^2}$$

三、角动量取向量子化

角动量量子数为 l 时, z 方向上的角动量有 $2l+1$ 个取值.

$$\vec{L} \cdot \vec{B} = L^2 \cos \theta \\ \Rightarrow \cos \theta = \frac{\vec{L} \cdot \vec{B}}{L^2}$$

§ 19 施特恩 - 格拉赫实验

不均匀磁场:

$$F = -\nabla(-\vec{\mu} \cdot \vec{B})$$

$$\hookrightarrow F_z = \mu_x \frac{\partial B_x}{\partial z} + \mu_y \frac{\partial B_y}{\partial z} + \mu_z \frac{\partial B_z}{\partial z}$$

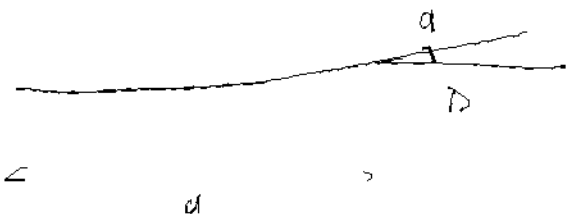
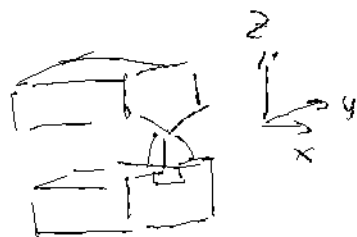
$$\hookrightarrow F_z = \mu_z \frac{\partial B_z}{\partial z} \quad \frac{\partial B_x}{\partial z} = \frac{\partial B_y}{\partial z} = 0$$

$$\left\{ \begin{array}{l} d = vt \\ z_1 = \frac{1}{2} \frac{F_z}{m} t^2 \end{array} \right. \quad \tan \alpha = \frac{F_z d}{mv} = \frac{F_z d}{mv^2}$$

$$z_2 = z_1 + D \tan \alpha = \frac{F_z d^2}{2mv^2} + D \frac{F_z d}{mv^2}$$

$$\propto \mu_z$$

$$\mu_z \uparrow \rightarrow z_2 \uparrow$$



对于角动量量子数为 L 的情况, $\mu_{L,z}$ 有 $2L+1$ 个取值.

但是, H, Li, Na, K, Cu, Ag 观察到偶数 g 的取值.

§ 20 电子自旋的假设

对电子: 角动量 $|\vec{S}| = \sqrt{S(S+1)} \hbar$, $S = \frac{1}{2} \rightarrow \mu_S = -|\vec{S}| \mu_B = -\frac{\sqrt{3}}{2} \mu_B$

z 方向: $S_z = m_S \hbar = \pm \frac{1}{2} \hbar$

$$\mu_{S,z} = m_S \mu_B = \pm \frac{1}{2} \mu_B$$

假设电子磁矩为经典理论的两倍: $g_S = 2$

← 与实验不符

$$\rightarrow \mu_S = -[S(S+1)] g_S \mu_B = -\sqrt{3} \mu_B$$

$$\mu_{S,z} = \pm m_S g_S \mu_B = \pm \mu_B$$

朗德 g 因子: = 实际 z 方向磁矩 (μ_B)
角动量在 z 方向投影 ($\hbar$)

$$\left\{ \begin{array}{l} \mu_J = -[J(J+1)] g_J \mu_B \\ \mu_{J,z} = -m_J g_J \mu_B \end{array} \right.$$

$$\mu_{J,z} = -m_J g_J \mu_B$$

轨道角动量: $J = L$, $g_L = 1$

自旋角动量: $J = S$, $g_S = 2$

单电子的g因子：外磁场不太强， $\vec{S}, \vec{L}$ 可耦合成 $\vec{J}$

总角动量： $\vec{J} = \vec{L} + \vec{S}$

$$\vec{\mu} = \vec{\mu}_L + \vec{\mu}_S \quad \hat{\mu}_J = \hat{\mu}_L + \hat{\mu}_S$$

$$\mu_J = \mu_J \cos\langle \vec{L}, \vec{J} \rangle + \mu_S \cos\langle \vec{S}, \vec{J} \rangle$$

$$= (-g_L \mu_B) \frac{\vec{J}^2 + \vec{L}^2 - \vec{S}^2}{2\vec{J}\vec{L}} + (-g_S \mu_B) \frac{\vec{J}^2 + \vec{S}^2 - \vec{L}^2}{2\vec{J}\vec{S}}$$

$$= -g_J \vec{J} \mu_B \quad \text{其中 } g_J = \frac{1}{2} \left(\frac{L^2 + S^2 + J^2}{L^2 S^2} \right)$$

$$\vec{J} = \vec{L} + \vec{S}, \quad S = \frac{1}{2}$$

$$\text{在 } z \text{ 方向, } S_z = \pm \frac{1}{2} \hbar, \quad L_z = l \hbar \quad J_z = L_z + S_z$$

$$\Rightarrow J = l + \frac{1}{2} \quad \text{或} \quad l - \frac{1}{2}$$

$$\begin{aligned} \Rightarrow g_J &= g_L \frac{\vec{J}^2 + \vec{L}^2 - \vec{S}^2}{2\vec{J}^2} + g_S \frac{\vec{J}^2 + \vec{S}^2 - \vec{L}^2}{2\vec{J}^2} \\ &= \frac{1}{2} (g_L + g_S) + \frac{1}{2} \left(\frac{L^2 - S^2}{J^2} \right) (g_S - g_L) \\ &= \frac{3}{2} + \frac{1}{2} \frac{S^2 - L^2}{J^2} \end{aligned}$$

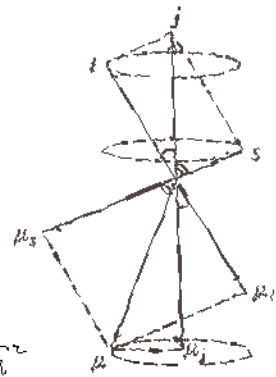
单电子原子的基态： $l=0$ $\textcircled{2}$ $S_{1/2}$ $l=0$ $J = \frac{1}{2} \rightarrow m_J = \pm \frac{1}{2}$

$$g_J = \frac{3}{2} + \frac{1}{2} \frac{S^2}{J^2} = 2$$

$$\mu_J = -m_J g_J \mu_B = \pm \mu_B$$

$$l=1 \quad {}^2P_{1/2}, {}^2P_{3/2}$$

$$l=2 \quad {}^2D_{3/2}, {}^2D_{5/2}$$



	l	s	j	g_j	m_j	$m_j g_j$
$^2S_{1/2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	2	$\pm \frac{1}{2}$	± 1
$^2P_{1/2}$	1	$\frac{1}{2}$	$1 - \frac{1}{2}$	$\frac{2}{3}$	$\pm \frac{1}{2}$	$\pm \frac{1}{3}$
$^2P_{3/2}$	1	$\frac{1}{2}$	$1 + \frac{1}{2}$	$\frac{4}{3}$	$\pm \frac{1}{2}, \pm \frac{3}{2}$	$\pm \frac{2}{3}, \pm 2$
$^2D_{3/2}$	2	$\frac{1}{2}$	$2 - \frac{1}{2}$	$\frac{4}{5}$	$\pm \frac{1}{2}, \pm \frac{3}{2}$	$\pm \frac{2}{5}, \pm \frac{6}{5}$
$^2D_{5/2}$	2	$\frac{1}{2}$	$2 + \frac{1}{2}$	$\frac{6}{5}$	$\pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}$	$\pm \frac{3}{5}, \pm \frac{9}{5}, \pm 3$

解法:

基态氢原子: $n=1, l=0, j=0+s=\frac{1}{2}, m_j=\pm\frac{1}{2}$

$$g_j = \frac{3}{2} + \frac{1}{4} \frac{\left[\frac{1}{2} \frac{3}{2} \right]}{\left[\frac{1}{2} \frac{1}{2} \right]} = 2$$

$$\mu_z = -m_j g_j \mu_B$$

$$= \pm \mu_B \rightarrow \text{双线}$$

§ 21 碱金属双线

主线系: $nP \rightarrow \infty S$

波数 λ 分裂间距 Δ

锐线系: $nS \rightarrow \infty P$

波数 λ 分裂间距

$\Rightarrow P$ 能级产生了分裂.

角动量 - 轨道相互作用

换到电子静止系:

$$\vec{L} = Z e \vec{v} = \frac{Z e V}{2 \pi r}$$

在中心 / 电子处:

$$B = \frac{\mu_0 I}{2 \pi r} = \frac{\mu_0 Z e V}{4 \pi^2 r^2} = \frac{1}{4 \pi \epsilon_0} \cdot \frac{Z e V}{c^2 r^2} \quad \mu_0 = \frac{1}{c^2 \epsilon_0}$$

$$\vec{B} = \frac{Z e}{4 \pi \epsilon_0} \frac{\vec{r} \times \vec{v}}{r^3} = \frac{Z e \vec{L}}{4 \pi \epsilon_0 E_0 r^3} \quad \vec{L} = \vec{r} \times m \vec{v}, \quad E_0 = m c^2$$

势能:

$$U' = - \mu_B \cdot \vec{B}$$

在核静止 / 实验室系中:

$$U = - \frac{1}{2} U' = - \frac{1}{2} \mu_B \cdot \vec{B}$$

$$= - \frac{Z e}{8 \pi \epsilon_0 E_0 r^3} \underbrace{\mu_B \cdot \vec{L}}_{\text{角动量 - 轨道相互作用}}$$

$$= \frac{Z e g_s \mu_B}{8 \pi \epsilon_0 \hbar E_0 r^3} \vec{S} \cdot \vec{L} = \frac{Z e^2}{8 \pi \epsilon_0 m^2 c^2 r^3} \vec{S} \cdot \vec{L}$$

估算: 氢原子的第二能级

$$R_2 = 1 \quad r = 2a_1 \quad s = 2a_2$$

$$U \approx 10^{-5} \text{ eV}$$

详细计算:

$$\vec{J} = \vec{S} + \vec{L} \quad , \quad \vec{J}^2 = \vec{S}^2 + \vec{L}^2 + 2\vec{S} \cdot \vec{L}$$

$$\vec{S} \cdot \vec{L} = \frac{1}{2} (\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$$

$$= \frac{1}{2} [J(J+1) - L(L+1) - S(S+1)] \hbar^2$$

对于单电子: $J = l \pm \frac{1}{2}$

$$\vec{S} \cdot \vec{L} = \frac{1}{2} l \hbar^2 \quad S \text{ 与 } L \text{ 同向}$$

$$= -\frac{1}{2} (l+1) \hbar^2 \quad S \text{ 与 } L \text{ 反向}$$

$$\langle \frac{1}{r^3} \rangle = \frac{Z^3}{n^3 l(l+\frac{1}{2})(l+1)a_1^3}$$

$$U = \frac{(a_1 Z)^4 E_0}{4n^3} \left[J(J+1) - L(L+1) - \frac{3}{4} \right] \frac{1}{l(l+\frac{1}{2})(l+1)}$$

$$U_{+1/2} = \frac{(a_1 Z)^4 E_0}{2n^3 (2l+1)(l+1)} \quad J = l + \frac{1}{2}$$

$$U_{-1/2} = \frac{(a_1 Z)^4 E_0}{2n^3 (2l+1)} \quad J = l - \frac{1}{2}$$

对氢原子的 $2p$ 分裂 $2p \rightarrow 2^2P_{1/2} \cdot 2^2P_{3/2}$

$$\Delta U = U_{+1/2} - U_{-1/2} \approx 4.53 \times 10^{-5} \text{ eV}$$

比较: 粗结构 $\alpha^2 E_0$

精细结构 $\alpha^4 E_0$

原子内部磁场:

$$U = -\vec{\mu} \cdot \vec{B}$$

$$\text{对单电子: } U = \pm \mu_B B, \quad \Delta U = 2\mu_B B = \frac{hc\alpha\lambda}{\lambda^2}$$

$$B = \frac{hc\alpha\lambda}{\lambda^2}$$

$$Z\mu_B\lambda^2$$

$$= 1857 \quad \text{鈉} \quad \lambda = 589 \text{ nm}, \quad \Delta\lambda = 0.6 \text{ nm}$$

碱金属

一、能级、量子数亏损

钻通贯穿效应：价电子可钻进原子核中

$$E = -\frac{1}{2} \mu \alpha^2 c^2 \frac{Z^{*2}}{n^2} \quad (Z^* > 1)$$

$$Z_{ns}^* > Z_{np}^* > Z_{np}^*$$

$$\text{令 } n^* = \frac{n}{Z^*} \quad \text{引} \times \text{量子数亏损: } \Delta n = n - n^*$$

二、超精细结构

$$\text{核自旋角动量: } I^2 = I(I+1) \hbar^2$$

$$\text{磁矩: } \mu_I = g_I \mu_N \frac{I}{\hbar}$$

$$\Delta E = -\mu_I \cdot B$$

$$= A \vec{I} \cdot \vec{J}$$

$$\text{总角动量 } F = \vec{I} + \vec{J}$$

$$F^2 = I^2 + J^2 + 2\vec{I} \cdot \vec{J}$$

$$\vec{I} \cdot \vec{J} = \frac{1}{2} (F^2 - I^2 - J^2)$$

$$\Delta E = A \vec{I} \cdot \vec{J} = g_I \frac{m_p}{m_p} m_p c^2 \alpha^2 \cdot \frac{1}{J(J+1)(I+1)} \frac{Z^2}{n^3} \cdot \frac{1}{2} [F(F+1) - I(I+1) - J(J+1)]$$

$$= \frac{A}{2} [F(F+1) - I(I+1) - J(J+1)]$$

例：氢原子基态 $1^2S_{1/2}$

$$J = \frac{1}{2}, \quad I = \frac{1}{2} \quad \rightarrow \quad F = 1, 0$$

$$E = \begin{cases} \frac{1}{4} a \\ -\frac{3}{4} a \end{cases}$$

§ 22 塞曼效应

1. 附加 \$z\$ 方向的磁场时:

$$U = -\vec{\mu} \cdot \vec{B} = -\mu_B B \quad \text{其中 } \mu_z = -m_j g_j \mu_B$$

$$\hookrightarrow U = m_j \mu_B B$$

$$E_1' = E_1 + U_1 = E_1 + m_1 g_1 \mu_B B \quad \text{其中 } m_1, m_2 \in [-m, m]$$

$$E_2' = E_2 + U_2 = E_2 + m_2 g_2 \mu_B B$$

$$h\nu' = E_2' - E_1'$$

$$= h\nu + (m_2 g_2 - m_1 g_1) \mu_B B$$

假设体系的角动量为 0, $g_1 = g_2 = 1$

$$h\nu' = h\nu + \Delta m \mu_B B \quad \text{其中 } \Delta m = 1, 0, -1, \quad h\nu = E_2 - E_1$$

$$\text{间隔频率 } \Delta\nu = \frac{\mu_B B}{h} = \frac{e\hbar}{2mc} = \frac{eB}{4\pi m} \quad \text{洛伦兹单位}$$

电子的轨道运动:

$$\vec{\mu} = -\frac{e}{2m} \vec{L}, \quad \frac{d\vec{\mu}}{dt} = \vec{\omega} \times \vec{\mu}$$

$$\text{其中 } \vec{\omega} = \frac{e}{2m} \vec{B} = \frac{eB}{2m}$$

$$\text{拉莫频率: } \nu = \frac{\omega}{2\pi} = \frac{eB}{4\pi m}$$

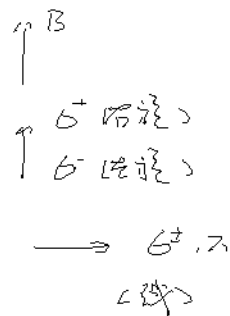
2. 偏振特性

原子和电子的角动量守恒:

$$\Delta m = 1 \quad \Delta L_z = -\hbar \rightarrow \Delta L_0 = \hbar \quad \sigma^+ \text{ 偏振.}$$

$$\Delta m = 0 \quad \Delta L_z = 0, \text{ 光子有固有角动量 } \hbar \quad \pi \text{ 偏振.}$$

$$\Delta m = -1 \quad \Delta L_z = \hbar \rightarrow \Delta L_0 = -\hbar \quad \sigma^- \text{ 偏振.}$$



3. 反常塞曼效应. (B较弱)

$$^2P_{1/2}, ^2P_{3/2} \rightarrow ^2S_{1/2}$$

外加磁场:

$$^2P_{1/2}: l=1, s=\frac{1}{2}, j=\frac{1}{2}, m_j = \pm\frac{1}{2}, g_j = \frac{3}{2} + \frac{1}{2} \frac{\frac{1}{2}(\frac{1}{2}+1) - 1 \cdot 2}{\frac{1}{2}(\frac{1}{2}+1)} = \frac{2}{3}$$

$$m_j g_j = \pm \frac{1}{3}$$

$$^2P_{3/2}: l=1, s=\frac{1}{2}, j=\frac{3}{2}, m_j = \pm\frac{1}{2}, \pm\frac{3}{2}, g_j = \frac{3}{2} + \frac{1}{2} \frac{\frac{3}{2}(\frac{3}{2}+1) - 1 \cdot 2}{\frac{3}{2}(\frac{3}{2}+1)} = \frac{8}{5}$$

$$m_j g_j = \pm \frac{2}{5}, \pm 2$$

$$^2S_{1/2}: l=0, s=\frac{1}{2}, j=\frac{1}{2}, m_j = \pm\frac{1}{2}, g_j = 2$$

$$m_j g_j = \pm 1$$

4. 帕邢 - 巴克效应. (B较强)

外场乃破坏了 $\vec{L} + \vec{S} = \vec{J}$, $\vec{L}$, $\vec{S}$ 分别进动, 不合成 J

强场判据: $\mu_B = \frac{e\hbar}{4\pi m_e} > \Delta E$ 精细结构 $\Delta E = h \delta \nu$

$$\begin{aligned} U &= -\vec{\mu} \cdot \vec{B} = -(\vec{\mu}_L + \vec{\mu}_S) \cdot \vec{B} \\ &= -\frac{e}{2m_e} (g_L \vec{L} + g_S \vec{S}) \cdot \vec{B} \\ &= -\frac{e\hbar}{2m_e} (2S_z + L_z) = -\frac{e\hbar}{2m_e} (2m_s + m_l) \end{aligned}$$

跃迁选择定则: $\Delta m_s = 0, \Delta m_l = 0, \pm 1$

例: 在帕邢 - 巴克效应的前提下, 考虑单电子的状态.

$$m_s = \pm\frac{1}{2}, m_l = 0, \pm 1$$

m_s	m_l	$2m_s + m_l$	$m_s m_l$	
$\frac{1}{2}$	1	2	+	----->
$\frac{1}{2}$	0	1	0	
$\frac{1}{2}$	-1	0	-	

$\frac{1}{2}$	-1	0	-
$-\frac{1}{2}$	1	0	-
$-\frac{1}{2}$	0	-1	0
$-\frac{1}{2}$	-1	-2	+

-----> 有旋-轨道耦合作用.

$$\propto m_s m_l$$

----->

5. 斯塔克效应 (E 较强)

$$U = -\vec{D} \cdot \vec{E} = -z D_z$$

$$D_z = -e \vec{r} z = -e \bar{z}$$

§1 × 量子数 n_1, n_2, m_l

$$n = n_1 + n_2 + |m_l| + 1, \quad (m_l \in [0, n-1])$$

$$\text{即 } n_2 = n_1 - m_l \in [-(n-1), (n-1)]$$

$$\text{斯塔克分裂: } \Delta E = 6.4 \times 10^{-5} \frac{E}{\text{\AA}} n n_2$$

§ 23 多原子光谱研究进展.